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MILITARY SPECIFICATION

REGULATOR, ENGINE GENERATOR: SOLID STATE, 28 VOLT, 25 AMPERE, DIRECT CURRENT

1. SCOPE

1.1 This specification covers a solid state regulator incorporating electronic devices to control reverse current and to regulate the voltage and current of the 28 volt, 25 ampere (AMP), direct current (DC) generator (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-E-489	-	Enamel; Alkyd, Gloss, (for Exterior and Interior Surfaces).
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.

Military

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	-	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-P-8585	-	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-B-11188	-	Batteries, Storage: Lead-Acid.
MIL-E-13856	-	Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL-F-13927	-	Fungus Resistance Test; Automotive Components.

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STANDARDS
Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1275	-	Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.

DRAWINGS

7355736	-	Generator, DC 28 Volt, 25 AMP.
10950808	-	Generator, DC 28 Volt, 25 AMP.
11631857	-	Regulator, Engine Generator, 28 Volt DC, 25 AMP Solid State.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. Preproduction samples of the regulator to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The preproduction samples shall be submitted to the Government for preproduction inspection to determine conformance to the quality assurance provisions of this specification (see 4.2). The preproduction samples submitted by a contractor shall be fully representative of the regulator to be supplied from production facilities and tooling.

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3.2 **Materials.** Material shall be as specified herein, in applicable specifications, and on applicable drawings. Materials not specifically designated shall be suitable for use in the regulator operating over an ambient temperature range from minus 65° to plus 225°F without any change in physical or dimensional properties that would result in operation of the regulator falling outside of the limits herein specified.

3.3 Design and construction. The regulator shall conform to Drawing 11631857.

3.3.1 Interchangeability. To provide for interchangeability, the regulator fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of Drawing 11631857. Electrical connectors, mounting dimensions, and location shall also conform to the detail requirements of the drawing.

3.3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.3.3 Standard parts. Military standard parts shall be used whenever possible.

3.3.4 Threaded surfaces. All screw threads shall conform to Handbook H28. The thread form, class, and the number of threads-per-inch shall be as specified on the applicable standards or drawings.

3.3.5 Polarity. The regulator shall be designed to operate with a negatively-grounded electrical system.

3.3.6 Accessibility. Access to internal components of the regulator shall be attained by removal of the cover. Two cover screws shall be sealed to prevent entry except by breaking the seals. Voltage and current regulation settings shall be non-adjustable or shall be locked to resist change when a torque of six inch-pounds is applied.

3.4 Electrical.

3.4.1 Reverse current. When tested as specified in 4.5.2.1, the reverse current shall be not more than five milliamperes.

3.4.2 Voltage regulation. When tested as specified in 4.5.2.2, the regulator shall establish and maintain output voltage at 28 ± 0.5 volts throughout the generator speed range of 2000-8000 revolutions per minute (RPM).

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3.4.3 Current regulation. When tested as specified in 4.5.2.3, the regulator shall maintain output current at 28.5 ± 1.5 AMPS.

3.4.4 Effective field circuit resistance. When tested as specified in 4.5.2.4, the regulator resistance at all times shall be not more than 2 ohms.

3.4.5 Field excitation. When tested as specified in 4.5.2.5, the regulator shall provide field excitation from a generator residual voltage of not more than 1.5 volts.

3.4.6 Load switching. During the test specified in 4.5.2.6, the regulator shall meet the abnormal system requirements of MIL-STD-1275, except that the steady state voltage max limit shall be 30 volts, the transient voltage max limit shall be 80 volts, and the peak to peak ripple voltage shall be not more than 5 volts. Subsequent to the test, the regulator shall meet the requirements of 3.4.1, 3.4.2, and 3.4.3.

3.4.7 High voltage transient. When tested as specified in 4.5.2.7, the regulator shall withstand the transient environment of MIL-STD-1275. Subsequent to the test, the regulator shall meet the requirements of 3.4.1, 3.4.2, and 3.4.3.

3.4.8 Reverse battery polarity. When tested as specified in 4.5.2.8, the regulator shall protect the generating system against damage by opening the circuit, and shall prevent reversal of generator polarity. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3 without servicing or replacement of any part.

3.5 Environmental.

3.5.1 High temperature (+ 225° + 5°F). When tested as specified in 4.5.3.1, the regulator shall meet the requirements of 3.4, except that the output voltage shall be 26.4 ± 0.8 and the output current shall be 22 ± 2 amperes.

3.5.2 Low temperature (-65° + 5°F). When tested as specified in 4.5.3.2, the regulator shall meet the requirements of 3.4, except that the output voltage shall be 29.6 ± 0.8 volts and the output current shall be 35 ± 3 amperes.

3.5.3 Vibration. When tested as specified in 4.5.3.3, the regulator shall evidence no breakage, loosening of parts, abrasion, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.4.1, 3.4.2, and 3.4.3.

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3.5.4 Shock. When tested as specified in 4.5.3.4, the regulator shall show no breakage, loosened or distorted parts, or other physical damage. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2, and 3.4.3.

3.5.5 Corrosion resistance. When tested as specified in 4.5.3.5, the regulator shall show no evidence of corrosion and shall subsequently meet the requirements of 3.4.1, 3.4.2, and 3.4.3.

3.5.6 Fungus and humidity. When tested as specified in 4.5.3.6, the regulator shall evidence no microbial growth or corrosion and shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.5.7 Waterproofness. When tested as specified in 4.5.3.7, the regulator shall show no evidence of leakage or damage and shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.5.8 Electromagnetic compatibility. When tested as specified in 4.5.3.8, the regulator shall provide electromagnetic interference reduction for the total generating system including the regulator, generator, and interconnecting shielded cable in accordance with the requirements of MIL-STD-461 for tactical vehicle components.

3.6 Endurance. When tested for 2000 hours as specified in 4.5.4, the regulator shall meet the requirements of vibration test method 201 of MIL-STD-202, 3.4.1, 3.4.2, 3.4.3, and 3.5.7. There shall be no servicing, adjustment, or replacement of parts throughout the test.

3.7 Finish.

3.7.1 Painting. All exterior surfaces of the regulator assembly shall be painted, except the electrical connections and mating surfaces on the mounting bracket. Corrosion resistant cover screws do not require painting.

3.7.2 Ferrous surfaces. Ferrous metal surfaces shall be cleaned in accordance with applicable method specified in TT-C-490. The cleaned surfaces shall be treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 to 1.25 mil thick or MIL-P-8585, 0.40 to 0.60 mil thick. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.7.3 Aluminum alloy surfaces. Aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541 or anodized in accordance with MIL-A-8625. Exterior surfaces shall be given one coat of primer conforming to MIL-P-8585, 0.40 to 0.60 mil thick. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

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3.7.4 Zinc and cadmium surfaces. Exterior zinc and cadmium surfaces shall be given one coat of primer conforming to MIL-P-8585, 0.40 to 0.60 mil thick. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.8 Identification of product.

3.8.1 Marking. The regulators shall be marked in accordance with MIL-STD-130.

3.8.2 Nameplate. A nameplate conforming to MIL-P-514 shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the plate shall include the following as a minimum:

Regulator,
25 AMPS, 28 Volts DC
Manufacturer's Identification
Manufacturer's Part Number
Serial Number
Military Part Number
Federal Stock Number
US

3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering, or other defects which affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. A preproduction sample of three regulators shall be submitted for preproduction inspection. Preproduction inspection shall be performed by the Government, at a place designated by the Government (see 6.2) and shall consist of performance of tests as specified in table I and examination for the defects specified in table II.

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Table I. Order of preproduction testing

Test	Paragraph No.	Specimens 1	2	3
Reverse battery polarity	4.5.2.8	X	X	X
High voltage transient	4.5.2.7	X	X	X
Field excitation	4.5.2.5	X	X	X
Load switching	4.5.2.6	X	X	X
Effective field circuit resistance	4.5.2.4	X	X	X
Reverse current	4.5.2.1	X	X	X
Voltage regulation	4.5.2.2	X	X	X
Current regulation	4.5.2.3	X	X	X
High temperature (+225° +5°F)	4.5.3.1	X	X	X
Low temperature (-65° +5°F)	4.5.3.2	X	X	X
Endurance	4.5.4	X		
Vibration	4.5.3.3		X	
Corrosion resistance	4.5.3.5		X	
Shock	4.5.3.4			X
Fungus resistance	4.5.3.6			X
Waterproofness	4.5.3.7.1		X	X
Electromagnetic compatibility	4.5.3.8		X	

4.2.1 Failure. Failure of a preproduction specimen to pass any specified examination or test may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all regulators from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with inspection level S-4 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination.

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4.3.2.1 Acceptable quality level. Each regulator selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects hall be classified as follows:

Table II. Classification of defects

	Major defects (1.0 AQL)	Examination Method
101	Incorrect dimensions affecting interchangeability (see 3.3)	SIE*
	Minor defects (2.5 AQL)	**Examination Method**
201	Incorrect dimensions not affecting interchangeability (see 3.3)	SIE*
202	Improper finish (see 3.7)	Visual
203	Improper identification of product (see 3.8)	Visual
204	Faulty workmanship (see 3.9)	Visual

*SIE - Standard inspection equipment.

4.3.3 Testing.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table III using an AQL of 2.5 percent defective.

Table III. Order of quality conformance testing

Paragraph	Test
4.5.2.1	Reverse current
4.5.2.2	Voltage regulation
4.5.2.3	Current regulation
4.5.3.7.2	Waterproofness, quality conformance

4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of one of each 500 units produced, except that not less than one nor more than four units shall be selected in any 30 day period. The control test sample shal

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selected from a lot which has passed the examination specified in 4.3.2 and the tests specified in table III, and shall be subjected to the tests listed in table IV in the order listed.

Table IV. Order of control testing

Paragraph	Test
4.5.2.8	Reverse battery polarity
4.5.2.7	High voltage transient
4.5.2.5	Field excitation
4.5.2.6	Load switching
4.5.2.4	Effective field circuit resistance
4.5.2.1	Reverse current
4.5.2.2	Voltage regulation
4.5.2.3	Current regulation
4.5.3.1	High temperature (+225°F ±5°F)
4.5.3.2	Low temperature (-65°F ±5°F)
4.5.3.7.1	Waterproofness, qualification and control

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause, at the option of the Government, to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, all tests shall be conducted at ambient room air temperature. The ambient room temperature shall be maintained at 77° ± 15°F. Unless otherwise specified herein, the test circuit shall include the regulator, battery, and generator. The generator employed in the test circuit shall conform to Drawing 7355736 or Drawing 10950808. The battery shall consist of two 12-volt batteries conforming to type 2HN of MIL-B-11188. The two batteries shall be connected in series. Batteries shall be fully charged and shall be kept at room temperature (see 6.3.2).

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table V and applicable test specification), test apparatus shall include chambers for conducting low temperature (-80°F), high temperature (+225°F) tests, and a submersion tank.

4.5.1.3 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification

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tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Standard measuring instrument tolerance accuracies are listed in table V.

Table V. Measuring instrument tolerances

Measuring instrument	Accuracy
Voltmeter DC	± 0.25 percent
Ammeter DC (Multi-Range)	± 0.50 percent
Tachometer	± 1 percent
Oscilloscope	± 3 percent

4.5.2 Electrical.

4.5.2.1 Reverse current. To determine conformance to 3.4.1, the operationally stabilized (see 6.3.1) regulator shall be connected in the test circuit with the generator operating at sufficient speed to deliver charging current to the battery. The generator speed shall then be gradually reduced to zero. The maximum value of the reverse current shall be observed on the ammeter.

4.5.2.2 Voltage regulation. To determine conformance to 3.4.2, the operationally stabilized regulator shall be connected in the test circuit. During the test, the resistance load shall be applied and removed suddenly. Obervations shall be made of the output voltage while the regulator is being subjected to the following test procedure:

(a) Increase generator speed from zero and operate generator at 2,000 RPM with resistance load at zero.
(b) With generator at 2,000 RPM, add resistance load to establish a total load of 20 AMPS.
(c) Increase generator speed to 5,000 RPM and maintain the 20 AMP load.
(d) With generator at 5,000 RPM, remove the resistance load.
(e) Increase generator speed to 8,000 RPM with resistance load at zero.
(f) With generator at 8,000 RPM, add resistance load to establish a total load of 20 AMPS.
(g) Decrease generator speed to 2,000 RPM and maintain the 20 AMP load.
(h) With generator at 2,000 RPM, remove the resistance load.
(j) Increase generator speed to 8,000 RPM with resistance load at zero.

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4.5.2.3 Current regulation. With the regulator operationally stabilized, the generator operating at 5,000 RPM, and the battery disconnected from the test circuit, the resistance load shall be adjusted to establish current regulation. The resistance load shall be further adjusted until the voltage output has been reduced from a maximum voltage (28 volts at -65°F, 27 volts at ∱ 77°F, and 25 volts at ∱ 225°F) to 6 volts. During the procedure the ammeter shall be observed to determine conformance to 3.4.3. Below 18 volts only the maximum current limit value shall apply (38 AMPS at -65°F, 30 AMPS at ∱ 77°F and 24 AMPS at ∱ 225°F). The procedure shall be repeated except that the battery shall be connected in the test circuit and the ammeter shall be observed at 25 volts.

4.5.2.4 Effective field circuit resistance. To determine conformance to 3.4.4, the operationally stabilized regulator shall be connected in the test circuit with battery disconnected. The generator speed shall be gradually increased from zero to produce a field current of 0.8 amperes in the unregulated mode and a load current adjusted to 18 amperes. The field current and generator-to-field voltage drop shall be recorded and the effective field circuit resistance calculated.

4.5.2.5 Field excitation. To determine conformance to 3.4.5, the regulator shall be connected in the test circuit and the generator speed gradually increased from zero until field excitation occurs. The generator voltage at excitation shall be recorded. The test shall be repeated, except that the battery shall be disconnected from the test circuit.

4.5.2.6 Load switching. To determine conformance to 3.4.6, the regulator shall be connected in the test circuit with the battery disconnected. With the generator speed at 5,000 RPM, the resistance load shall be adjusted to deliver a current regulated load at 28 volts at -65°F, 27 volts at ∱ 77°F and 25 volts at ∱ 225°F. The load shall be switched on for one second, off for one second, and repeated for a period of fifteen minutes. Load switching time shall be 1.5 ∱ 0.5 milliseconds. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.2.7 High voltage transient. To determine conformance to 3.4.7, the regulator shall be connected in the test circuit with the battery disconnected and the generating system inoperative. A negative voltage pulse of 100 volts, square wave, of 50 millisecond duration, shall be applied to the regulator output terminal, once every two seconds, equally spaced, for a period of ten minutes. The test shall then be repeated, except that the voltage pulse shall be positive and the period shall be five minutes. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2 and 4.5.2.3.

4.5.2.8 Reverse battery polarity. To determine conformance to 3.4.8, the regulator shall be connected in the test circuit. With the generator inoperative, the battery polarity shall be reversed and a voltage applied in

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increments from maximum voltage to a minimum of 6 volts. Then with maximum voltage applied, the generator shall be operated. During the tests, checks for open circuit shall be made. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.3 Environmental.

4.5.3.1 High temperature (225° ± 5°F). To determine conformance to 3.5.1, the regulator shall be stored for 24 hours in an ambient temperature of 225° ± 5°F. With ambient temperature maintained at 225° ± 5°F, the regulator shall be tested in accordance with 4.5.2.8, 4.5.2.7, 4.5.2.5, and 4.5.2.6. While remaining in an ambient temperature of 225° ± 5° and connected in the test circuit, the regulator shall function continuously with field current of .75 amperes for one-half hour at each of the following conditions:

a. Output voltage of 25 volts in the current regulated mode.
b. Load current of 12.5 AMPS in the voltage regulated mode.

The regulator shall than be tested in the following order in accordance with 4.5.2.4, 4.5.2.1, 4.5.2.2, and 4.5.2.3, except that the temperature shall be 225° ± 5°F.

4.5.3.2 Low temperature (-65° ± 5°F). To determine conformance to 3.5.2, the regulator shall be stored for 24 hours in an ambient temperature of minus 80° ± 5°F. The regulator shall then be tested in the following order in an ambient temperature of minus 65° ± 5°F in accordance with 4.5.2.8, 4.5.2.7, 4.5.2.5, 4.5.2.6 5.5.2.4, 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.3.3 Vibration. To determine conformance to 3.5.3, the regulator mounted as in intended use shall be subjected to test condition A, method 204 of MIL-STD-202, except that the sweep time cycle shall be performed 24 times. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.3.4 Shock. To determine conformance to 3.5.4, the regulator shall be subjected to test condition A, method 213 of MIL-STD-202. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.3.5 Corrosion resistance. To determine conformance to 3.5.5, the regulator shall be subjected to the salt spray corrosion resistance test specified in method 101 of MIL-STD-202. Regulators shall be tested for 400 hours sealed, and internal components shall not be exposed. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

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4.5.3.6 Fungus and humidity resistance. To determine conformance to 3.5.6, the regulator shall be subjected to the fungus resistance test in accordance with class 1 of MIL-F-13927, except that the period of exposure shall be a continuous 90 days. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.3.7 Waterproofness.

4.5.3.7.1 Qualification and control. For qualification and control test purposes, regulators shall be subjected to the waterproofness test specified in MIL-E-13856 for type II, class 2 components to determine conformance to 3.5.7. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2, and 4.5.2.3. Electrical operation during submersion is not required.

4.5.3.7.2 Quality conformance. For quality conformance test purposes, regulators shall be subjected to the acceptance test inspection specified in MIL-E-13856 for type II class 2 components, except that tap water shall be used instead of salt water solution, to determine conformance to 3.5.7.

4.5.3.8 Electromagnetic compatibility. To determine conformance to 3.5.8, the regulator shall be tested in accordance with MIL-STD-462.

4.5.4 Endurance. To determine conformance to 3.6, the regulator shall be connected in the test circuit and subjected to the following procedure. The regulator shall function continuously when cycled (see figure 1) at 24 cycles per hour for 2000 hours. Each 1000 hour test period shall be divided into time-temperature periods; and each time-temperature period shall be divided between different resistance loads as specified in table VI. After each 1000 hour test period, the regulator shall be subjected to the vibration test specified in method 201 of MIL-STD-202 and tests specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.3.7.1.

Table VI. Endurance test sequence

Time	Temperature	Resistance load	
		one-half of time-temperature period	one-half of time-temperature period
Hours	Fahrenheit	Amperes	Amperes
190	225°	18	4
190	77°	23.5	4
50	-65°	29	4
140	0°	27	4
190	77°	23.5	4
240	160°	20	4

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4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The regulators covered by this specification are intended for use in 28 volt DC nominal electrical systems equipped with charging generators. The regulators are intended for use on tactical and combat (military) vehicles and industrial applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Military part number.
(c) Other applicable standards or drawings if required (see 3.2).
(d) Treatment and painting if other than specified (see 3.7).
(e) Where preproduction samples are to be sent (see 4.2).
(f) Selection of applicable packaging standard or packaging data sheet (see 5.1).

6.3 Definitions.

6.3.1 Operationally stabilized. An operationally stabilized regulator is defined as a regulator which has operated for one hour with the generator speed adjusted to 3000 RPM and the load adjusted to 12.5 AMPS.

6.3.2 Fully charged battery. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than two amperes at 28 volts at room temperature ($77^{\circ} \pm 15^{\circ}F$).

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2920-A340

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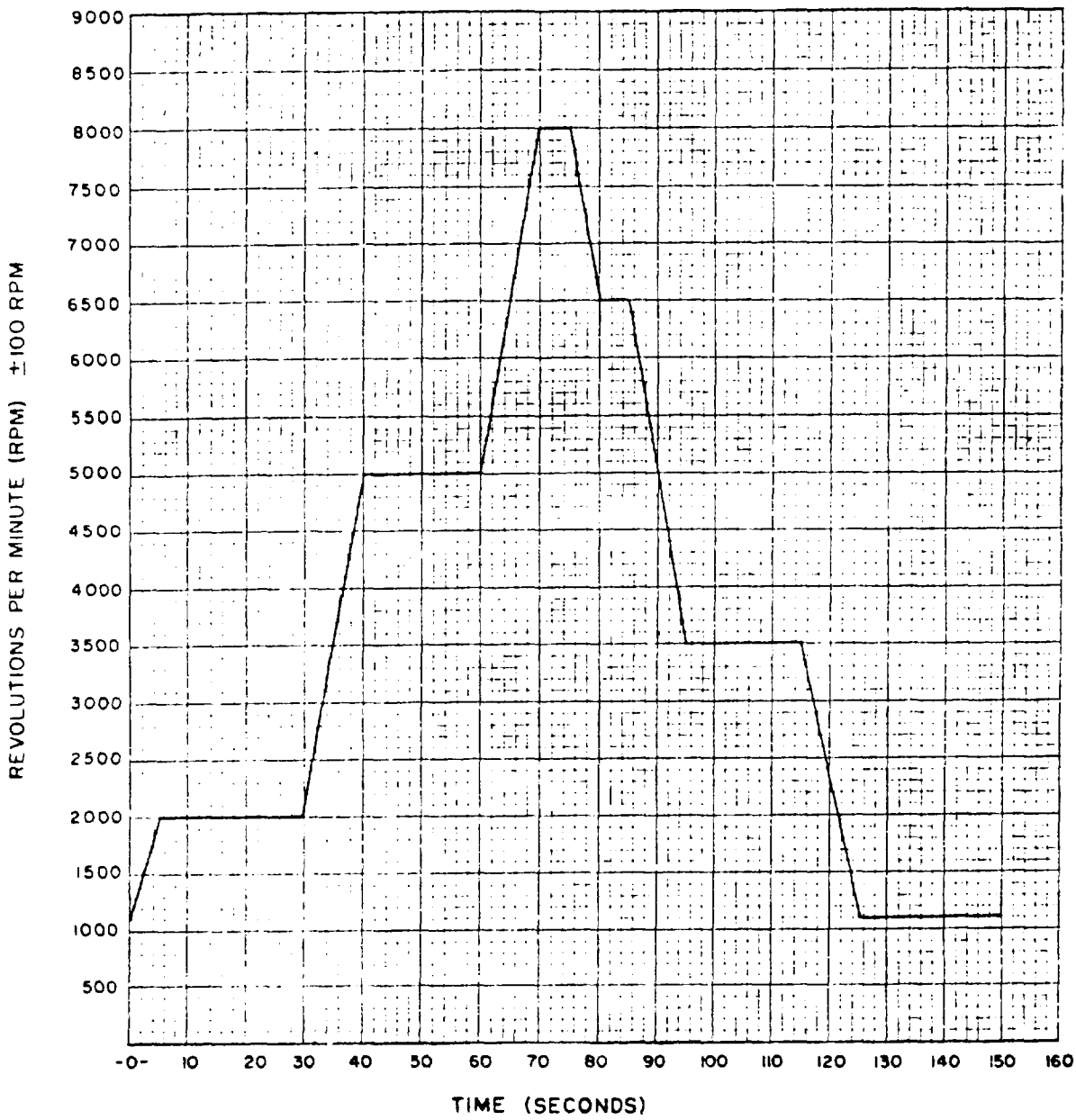


FIGURE 1.

ENDURANCE CYCLE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-62067B(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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